

then audibly suggests that the driver pull over and take rest.

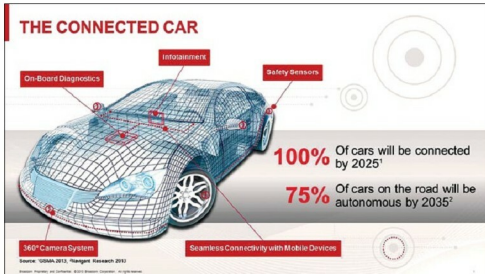
Ford has an interesting emergency assist feature that connects vehicle occupants with emergency services in the misfortune of an accident, providing vehicle location and an open line for communication. "Ford India is the first auto manufacturer and EcoSport is the only vehicle in its segment to offer a potentially life-saving emergency assistance system. This is a vehicle-based, no-cost, non-subscription call-for-help system that is already available in more than five million Ford vehicles globally. The service is provided free of charge for the ownership cycle of the vehicle," says Vinay Piprasania, executive director, Marketing, Sales and Service, Ford India.

"The ability to determine an available parking space along a roadway, and back in automatically, with no more effort from the driver than the push of a button is being demonstrated by OEMs today," says Tim Lau, associate product line director, Automotive, Broadcom.

Mahindra Reva's new cars feature another interesting aspect called Feedback Orientation. "Since the car is not an isolated object but a part of a system, we are able to get feedback on its performance. The car is able to tell the driver how he is driving, and compare the performances over a period of time, so that the driver can tweak his driving style for efficiency," says Chetan Maini, chief executive officer, Mahindra Reva Electric Vehicles Pvt Ltd.

As ADAS gets more advanced and reliable, it has started to induce some amount of autonomy in cars. Whilst it is still not possible to put a car on auto-drive mode and sit back, next-gen cars will surely let the driver relax and chat while driving. "Automated driving technology, being demonstrated now, will be available as an option on actual models within a few years. It takes control of cars for brief periods.

The technology, which uses a series of camera, radar, laser and ultrasonic sensors 'to become aware of its surroundings,' will relieve drivers of certain tasks—such as keeping a car in lane while rolling down a highway at



The connected car (Courtesy: Broadcom)

a steady speed. Ultimately, the technology could take over for the driver in stop-and-go traffic including total control of steering, braking and accelerating. Industry analysts have forecast that 75 per cent of cars on the road will be autonomous by 2035," explains Lau.

More updates.

► Audi's new Traffic Light Assist system combines the car's GPS navigation system with a city's traffic light information system to let the driver know in advance simply how much time he has to zip past an approaching signal on his route before it turns red. The car maker is testing the system in certain developed nations.

► BMW aims to bring autonomous driving assistance into production by end of the decade. In January, they demonstrated a car with advanced computing and GPS technology used in guided missile systems, which is capable of freeing the driver by operating the accelerator, steering and brakes. Its intelligence, supported by a new ultrasonic radar and 360-degree stereo camera technology, enables the car to change lanes to overtake slower vehicles and then pull back in without any assistance from the driver.

My car, my connected world

Now everyone wants to add connectivity to the car to be able to support features such as a Wi-Fi hotspot, greater amounts of infotainment endpoints,

infotainment data, more sensors and cameras, scalable in-vehicle networks, car-to-car communications and other types of applications, but what is the best technology to achieve this with best bandwidth and least cost?

Some of the options are CAN and LIN with a throughput in the range of 2kbps; FlexRay, which does about 10Mbps; and MOST, which offers 150 megabits of shared network bandwidth. But companies like Broadcom bet big on Ethernet. "Ethernet solves a real problem here in that we can address very high bandwidth network applications in a very cost-effective way, over a single pair unshielded twisted cable. It allows the automotive OEMs and Tier-1's to implement very high-end features but at a very competitive price point," adds Lau.

Ethernet also offers benefits like ease of integration and availability from a standard, open and widely-deployed technology. Standardisation is essential as a major enabler for new and innovative in-vehicle applications, allowing automotive manufacturers to meet customer expectations and keep the bottom line in check. Standards-based solutions not only reduce time-to-market, they also ensure availability, lifecycle, upgradability and interoperability.

"Ethernet is ubiquitous in the consumer, enterprise and service provider spaces and yet it has never really been used in an automotive environment, except for one very niche application, which is on-board diagnostics (OBD).